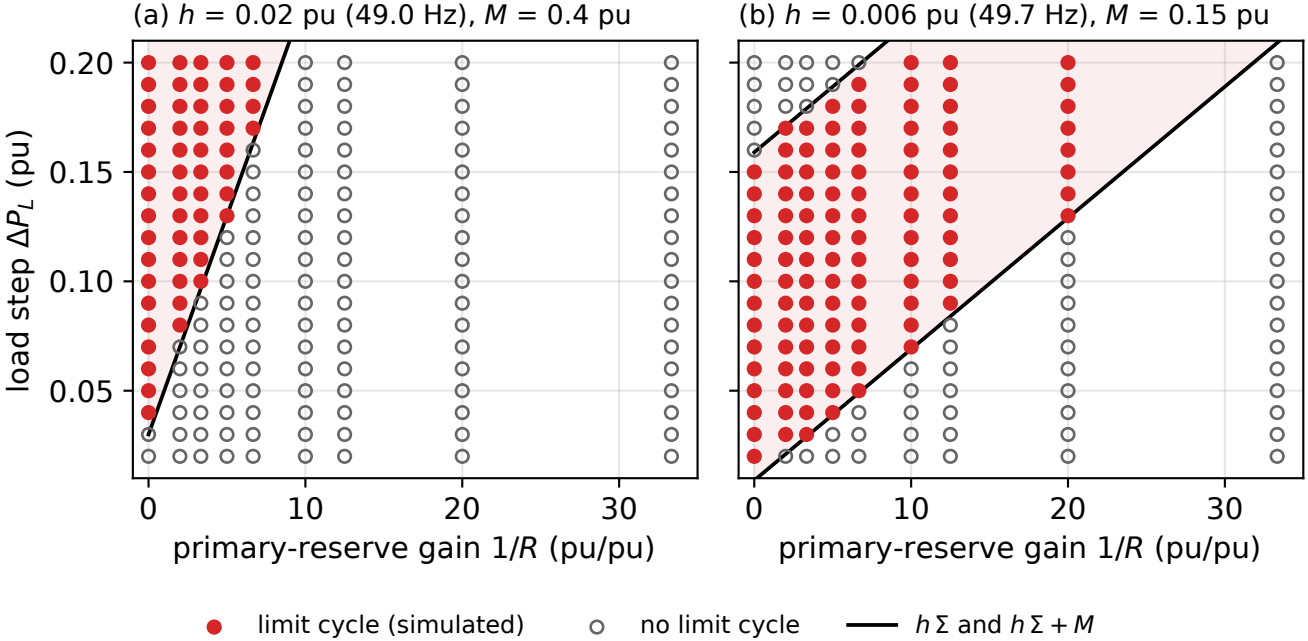
(a) h = 0.02 pu (49.0 Hz), M = 0.4 pu
(b) h = 0.006 pu (49.7 Hz), M = 0.15 pu
load step ΔP_L (pu)
0.20
0.15
0.10
0.05
0
10
20
30
primary-reserve gain 1/R (pu/pu)
limit cycle (simulated)
no limit cycle
$h\,\Sigma$ and $h\,\Sigma + M$